UNIVERSITY OF THE P...

BS (After Associate Degree) : Fifth Semester – Fall 2021

Time: 3 Hrs. Ma

Paper: Computational Physics – I Course Code: PHYS-3502

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDE

(6x5=30)

Q.1. Solve the following questions.

I. Explain with example the following terms in C++:
   a. float x
   b. switch()
   c. sqrt()

II. Write syntax with example for the followings in C++:
   a. do-while loop
   b. for loop

III. Write a program to print the given series and sum of the series: , ask user to input value of x.

IV. Write C++ program with function to calculate $f(x) = 20x^2 - 8x + 3$ and print values for x and f(x). Implement for seven iterations.

V. Write C++ program to calculate and print factorial of a number taken from user using while loop. Implement for six iterations.

VI. Write C++ program to calculate and print equivalent resistor of seven resistors connected in series.

Q.2. Solve the following questions. (3x10=30)

I. Write a C++ program for the decay of current in a simple RL-circuit using Euler's method with initial conditions: $r=10\Omega$, $L=5H$, initial time 0, time step 0.1, maximum time 2.5sec., initial current 5A and voltage $v=0$ volts. Print current against time values. How you can convert the same program for the growth of current in the circuit?

II. Write a program to evaluate the integral by Trapezoidal rule, I= 33.604.

III. Write a program to find a product of two matrices of order 3x3 and calculate the trace of these matrices

Fifth Semester-Fall 2021

Question # 01:

(I) (a):

float x :

In C++, float is the data type that represents a floating point numbers. The floatpoint numbers are used to represent very large or small values / numbers with a fractional part.

e.g:

```
#include <iostream.h>
#include <conio.h>
void main()
{
float x = 3.14;
//declare and inatialize a float
variable
cout<< "The value of x is:" <<endl;
cin>> x;
```

```
x = x * 2.0;
//perform an operation on float variable
Cout<< "The new value of x is:"<<en
getch();
}
```

(b) switch():

In C++ program, the switch statement is a control structure that allows you to execute different blocks of code based on the value of an expression. It's a short-cut for if-else series.

Basic syntax:

```
switch (Expression)
{
Case value 1:
// code to execute if expression equal to value 1.
break;
```

```
Case value 2:
// code to execute if value 2
equal to expression
  break;
... .. .
default;
 // code to execute if
 expression doesn't match
 any case
  break;
 }
}
```

(c) sqrt():

In C++, the sqrt() function is used to calculate the square root of a number. It's a mathematical function that returns the positive square root of a given value.

e.g:

```
# include <iostream.h>
# include <conio.h>
```

```
void main()
{
    double num = 16.0;
    double root = sqrt(num);
// Calculate the sqrt of number
    cout << "the square root is:" << num <<
    endl;

    getch();
}
```

II

(a) do-while loop:

The do-while loop is a type of loop in C++ that executes a block of repeatedly while a certain condition is true. It is similar to while loop but with key difference.

e.g:

```
# include <iostream.h>
# include <conio.h>
```

```
Void main()
{
  int i = 0;
  do
  {
    cout << "Hello, World!"<<i <<
    endl;
    i++;
  }
  while (i<5);
  getch();
}
```

(b) for loop:

The for loop is a type of loop in C++ that allow you to execute block of code repeatly for a specific number of iterations.

Syntax:

for(initilization; condition; increment/decrement)

```
# include<iostream.h>
# include <conio.h>
void main()
{
for (int i=0; i<5; i++)
{
cout<< "Hello, World!"<<i<<endl;
}
getch();
}
```

(III)

```
# include <iostream.h>
# include <conio.h>
void main()
{
clrscr();
int x,n;
double sum = 0;
cout<< "Enter the value of x:";
cin>> x;
cout<< "Enter the value of n:";
```


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```
cin>> n;
cout<< "series:";
for( int i= 1; i<=n; i++)
  {
    cout << "sum of series:"<< endl
       sum = sum+ x/i;
    }
   cout<< "Sum of series": << sum<<
     getch();
      }
```

(IV)

```
# include<iostream.h>
# include <conio.h>
void main()
 {
     float fx;
      float x[7];
    for  int(i=0; i<7; i++)
    {
 cout<<"Enter the value of x=";
  cin>> x[i];
```

```
f(x) = 20*x[i]*x[i]-8*x[i]+3;
cout<< "f(x)=" << \t << fx <<endl;
getch();
}
```

(V)

```
#include<iostream.h>
#include<conio.h>
#include<math.h>
void main()
{
int n;
int fact=1;
int i=1;
cout<<"Enter the the number"<<endl;
cin>>n;
while(i<=6)
{
fact=fact*i;
i=i+1;
}
```

```
cout<<"The factorial of number=";<<
  fact<<endl;
  getch ();
  }
```

(VI)

```
# include <iostream.h>
# include <conio.h>
# include <math.h>
void main()
  {
    double r1, r2, r3, r4, r5, r6, r7,
     equivalent_resistance;
   cout<<"Enter the value of resistor 1
   (in ohms):";
    cin>> r1;
   cout<<"Enter r2(in ohms):";
   cin>> r2;
   cout<<"Enter r3 (in ohms):";
   cin>>r3;
   cout<<"Enter r4:";
```

```
cin>> r4;
cout<<"Enter r5:";
cin >> r5;
cout << " Enter r6":;
cin>> r6;
cout << " Enter r7";
cin>> r7;
equivalent-resistance = r1+r2+r3+
r4+r5+r6+r7;
cout<< "Enter equailent resistance
of seven resistors connected in
series =" << equivlent-resistance <<
"ohms" << endl;
getch();
}
```

Long Questions:

(1)

```
#include <iostream.h>
#include <conio.h>
void main()
{
```

// $\frac{di}{dt}$ = rate of change of current w.r.t time

```
cout << "Decay of current is RL-series
    circuit" << endl;
float i = 5, V=0, h = 0.1;
float tmax = 2.5 , L = 5, t=0, r=10
float ia [1000], ta[1000]; didt ;
  int ind = 0;
while (t <= tmax)
{
didt = (V - i*R)/L ;
  ta [ind] = t;
  ia [ind] = i;
i = i+ didt * h;
   t = t+h ;
   ind = ind+1;
   }
for (x=0, x<ind; x++)
  {
  cout << ta [x] << "\t" << ia [x] <<
  "\t" << endl;
   }
  getch ();
   }
```

Also explain how you convert this program for growth of current in circuit?

```
# include <iostream.h>
# include <conio.h>
void main()
 { cout << "Growth of current in RL-series";
 float t=0, h=0.1; V=0 ; i=5;
 float tmax= 2.5; L= 5; r=10;
 float ia[1000], ta[1000]; didt ;
 int ind =0;
 while (t <= tmax)
 {
  didt=(V-i*R)/L;      :: didt= di/dt
  ta[ind]= t;
  ia[ind]= i;
  i= i+ didt*h;
  t = t+h;
  ind = ind+1 ;
 }
 cout<<t<<"\t"<<i<<endl;
```

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```
for(x=0; x<ind, x++)
 cout << ta[x] << "\t" << ia[x] << endll;
   getch();
      }
```

(II)

```
float f(float x)
{
   float b;
   b=x*x;
   returnb;
   }
```

(II)

```
# include <iostream.h>
# include <conio.h>
# include <math.h>
[illegible]
     float f(float x)
        {
          float b;
           b=(x*x);
```

```
returnb;
}
void main()
{
cout << "Trapezoidal rule"<<endl;
float h, n, low, up, sum=0;
cout << "Enter lower limit="<<endl;
cin >> low;
cout << "Enter upper limit="<<endl;
cin>>up;
cout << "Enter num of intervals:"<<endl;
cin>>n;
h=(up-low)/n;
cout >>"Enter step size="<< endl;
cin >> n;
Sum = f(low)+f(up);
for (int i=1; i<n; i++)
{
Sum+= 2*f(low+i*h);
Sum= sum*h/2;
}
cout <<"I=" << sum << endl;
```

```
getch();
}
```

(III)

```
#include <iostream.h>
#include <conio.h>
#include <math.h>
void main()
{
int i,j,k;
int a[3][3],b[3][3],pro[3][3];
// matrix 1 = a , matrix 2 = b
cout<<"Enter elements of matrix1:";
for(i=0 ; i<3 ; i++)
{
for(j=0;j<3;j++)
{
cin>> a[i][j];
}
}
cout<<"Enter elements of matrix2:";
for( i=0; i<3; i++)
```


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```
{
  for (j=0; j<3; j++)
   {
    cin>> b[i][j];
     }
      }

// Matrix Multiplication
for (i=0; i<3; i++)
  {
   for(j=0; j<3; j++)
     {
       product [i][j] = 0;
   for (k=0; k<3; k++)
   {
   product [i][j] += a[i][k] * b[j][k];
     }
       }
        }
  cout << "Product of Matrices": ;
       }
    cout << endl;
```


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```
// Calculate trace of matrices.
int trace 1 = 0, trace 2 = 0;
for (i=0; i<3 ; i++)
 {
  trace 1 += a[i][i];
  trace 2 += b[i][i];
  }
 cout << "Trace of Matrix 1:" <<
  trace 1 << endl;
 cout << "Trace of Matrix 2:" <<
  trace 2 << endl;
  getch();
   }
```